Quick start: Ames_sub1

Run a complete SWAT+ simulation in a few minutes using the small reference dataset shipped with the source repository.

What it is

`Ames_sub1` is a twelve-HRU dataset near Ames, Iowa. It simulates 1975 through 2020 on a daily time step (see `time.sim`). The dataset is in the SWAT+ source tree at `refdata/Ames_sub1`.

Prerequisites

- A SWAT+ executable. See Install.
- A copy of the `Ames_sub1` folder. If you cloned the source repository, it is at `swatplus/refdata/Ames_sub1`. Otherwise, download it from the link above.

Run it

1. Copy the executable into the `Ames_sub1` folder, or note its full path.
2. From a terminal, change into the `Ames_sub1` folder:

   ```
   cd path/to/Ames_sub1
   ```

3. Run the model:

   ```
   # Linux / macOS
   ./swatplus-3.0.4-gnu-lin_x86_64

   # Windows
   swatplus-3.0.4-gnu-win_amd64.exe
   ```

A simulation of this size takes a few seconds to a minute depending on your machine.

What you should see

While running, SWAT+ writes progress to the console and to `simulation.out`. When it finishes successfully, the last lines of `simulation.out` contain:

```
 Execution successfully completed
```

The same line is written to `success.fin`.

The folder now contains a set of output files. With the default `print.prt` for this dataset, every `objects` row is set to `avann = y`, so each enabled object group writes an annual-average file. The annual-average files that appear in `refdata/Ames_sub1` after a successful run include:

```
basin_wb_aa.txt          basin water balance
basin_nb_aa.txt          basin nutrient balance
basin_ls_aa.txt          basin landscape (sediment)
basin_pw_aa.txt          basin plant and water stress
basin_aqu_aa.txt         basin aquifer
basin_cha_aa.txt         basin channel (legacy routing)
basin_sd_cha_aa.txt      basin channel (SWAT-DEG routing)
```

`basin_res_aa.txt`	basin reservoir
`basin_psc_aa.txt`	basin point-source / recall
`basin_crop_yld_aa.txt`	basin crop yields
`crop_yld_aa.txt`	crop yields (HRU detail)
`hru_wb_aa.txt`	HRU water balance
`hru_nb_aa.txt`	HRU nutrient balance
`hru_ls_aa.txt`	HRU landscape balance
`hru_pw_aa.txt`	HRU plant and water stress

Open `basin_wb_aa.txt` to see the simulated water balance. The columns are documented in Water balance outputs.

If it fails

- **`Error opening file...`** Check that you are running the executable from inside the `Ames_sub1` folder, not from its parent.
- **`Permission denied`** on Linux or macOS. Run `chmod +x` on the executable.
- The model exits before writing output. Check `diagnostics.out` and `checker.out` for the first error.

For other errors, see FAQ and troubleshooting.

Next

- Read Project folder anatomy to learn what each file in `Ames_sub1` does.
- Walk through the dataset slowly in the Ames_sub1 walkthrough.
- Build your own watershed in Your first watershed.